

The model is the leak.

Pseudonymisation removes the name from the record.

We proved that the trained model still gives the patient away.

Repository: <https://github.com/waheedaiman/Privacy-Preserving-Analytics-on-Sensitive-Government-Data>

Team **unlisted**

6,000

patients, three times our Stage 2 cohort

3 runs

every model trained and attacked three times

0.6%

accuracy we pay for a formal privacy guarantee

The law moves the model, not the data. We assumed that was safe.

The records cannot travel

Federal Law No. 2 of 2019. UAE health data must not leave the State.

So we send the trained model instead.

The names are already gone

PDPL Art. 20 asks for pseudonymisation.

No name, no address, no file number ever leaves the hospital.

So we asked one question

Can an outsider tell whether YOUR record was used to train this model?

If the answer is yes, the model has disclosed that you were a patient at that hospital, in that period. No diagnosis required.

Our job was not to claim that this gap exists. It was to measure it.

Records stay home. Only protected maths travels.

Hospital A

3,000 patients. Trains locally, then adds the noise.



Aggregation server

Averages the protected updates.
Never sees a record.



Hospital B

3,000 patients. Trains locally, then adds the noise.

Eight rounds. The protected update goes up, the averaged model comes back. The records never move.

The server is not trusted

Noise is added inside the hospital.

A stolen server learns no more than the epsilon bound allows.

The budget does not add up

No patient is at both hospitals.

The cost is the larger of the two, not the sum.

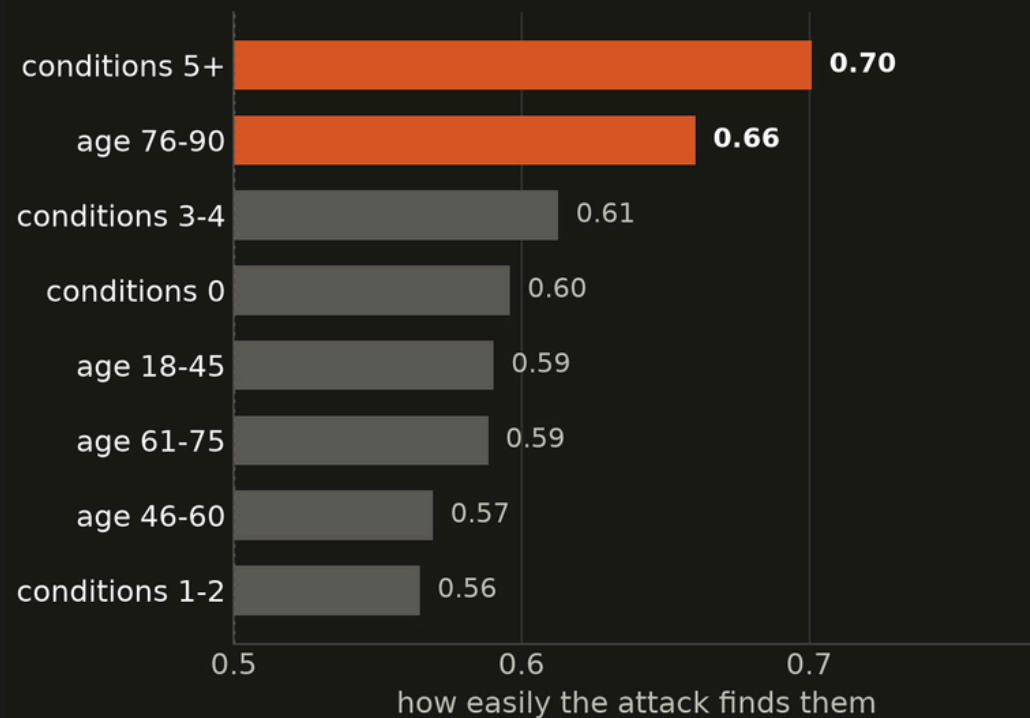
Scale makes privacy cheap

In Stage 2, at 2,000 patients, privacy cost 2.5% of the accuracy.

At 6,000 patients it costs 0.6%.

More data does not make privacy more expensive. It makes it cheaper.

We attacked ourselves. Who got hurt?



Careless model. 0.5 means the attacker is guessing.

0.586

against the careless model. Our audit certifies a real leak at epsilon 2.36.

0.506

against the DP model. Chance level in all 3 runs, spread 0.007.

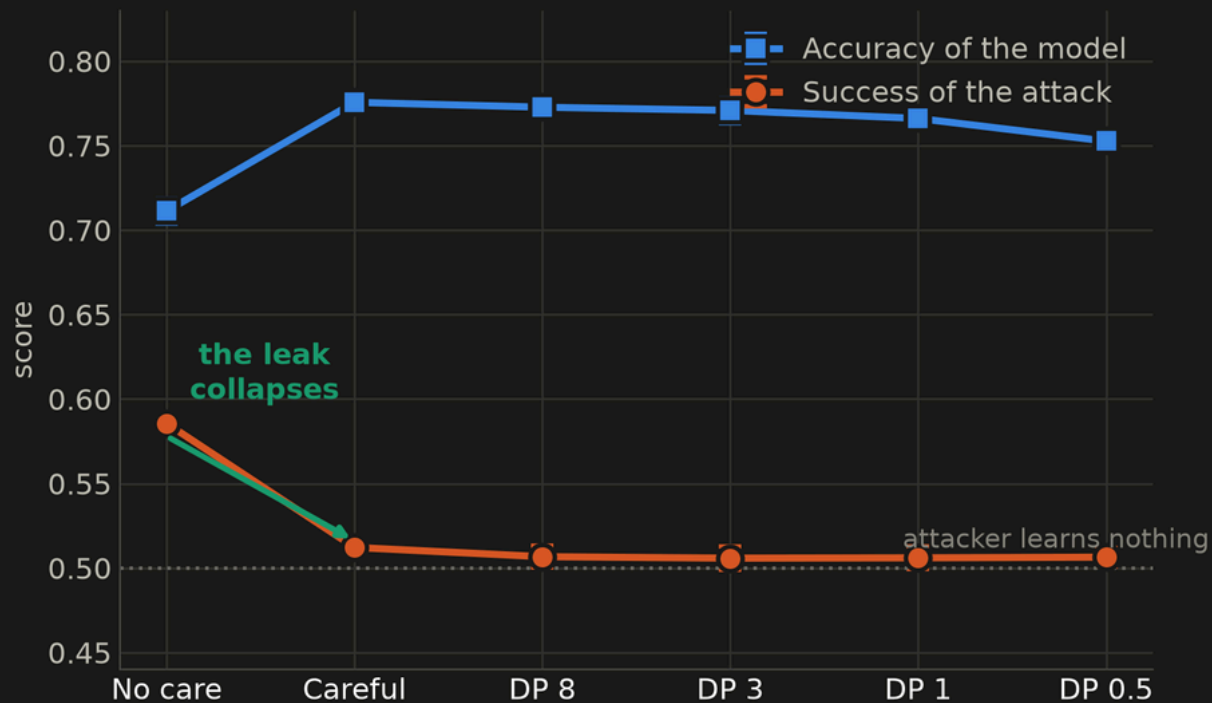
Seven attacks, including the state of the art

LiRA builds 96 shadow models, so it knows what a normal score looks like for every single patient.

At a 1% false alarm rate it finds 1.7 times more people than the classic attack.

The attack is best at finding the patients with the most to lose.

A formal guarantee for 0.6% of the accuracy.



Mean of 3 runs. The best score any model could reach is 0.784.

Ship epsilon 3

epsilon 2.99, delta 0.00001, for each hospital. Accuracy 0.771 against 0.776 with no privacy at all.

Careless is not cheap

The leaking model was also the worst, at 90.7% of the best score.

Care is not a guarantee

LiRA still scores 0.512 on the careful model. Only DP reaches chance.

Scale makes privacy cheap

Three times the patients cut the cost from 2.5% to 0.6%.